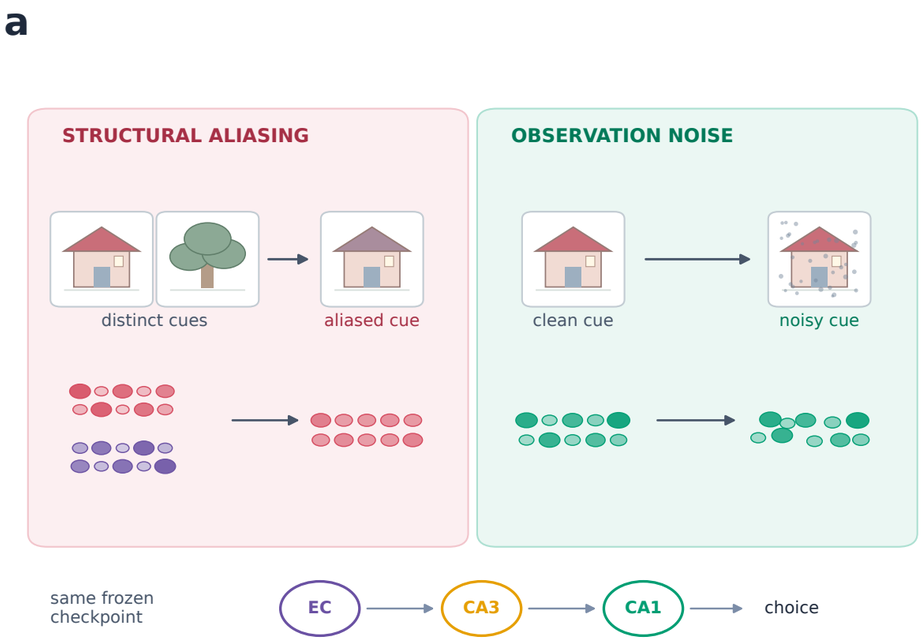
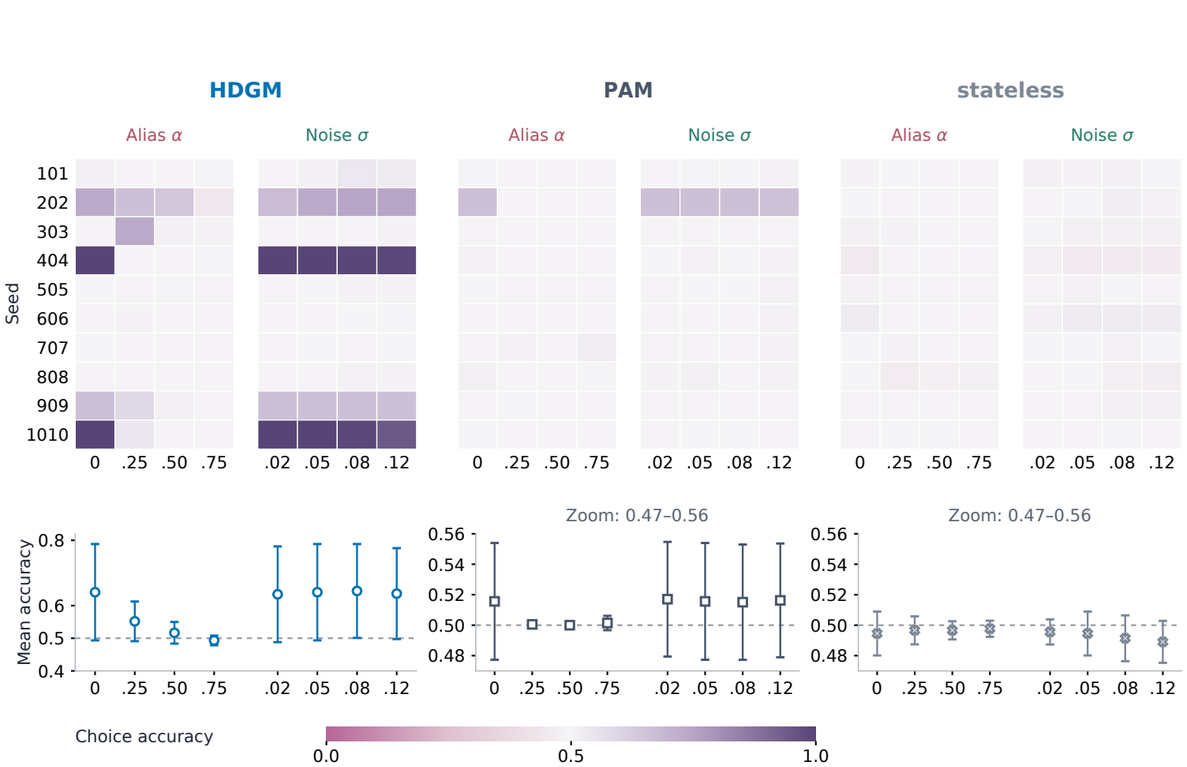
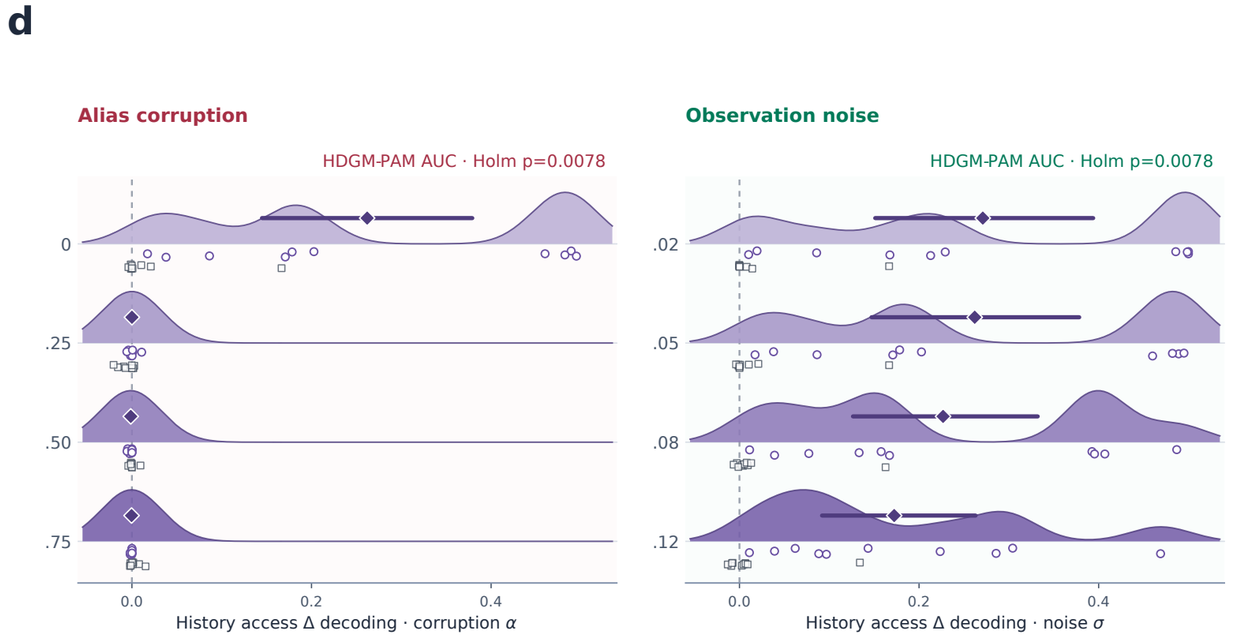
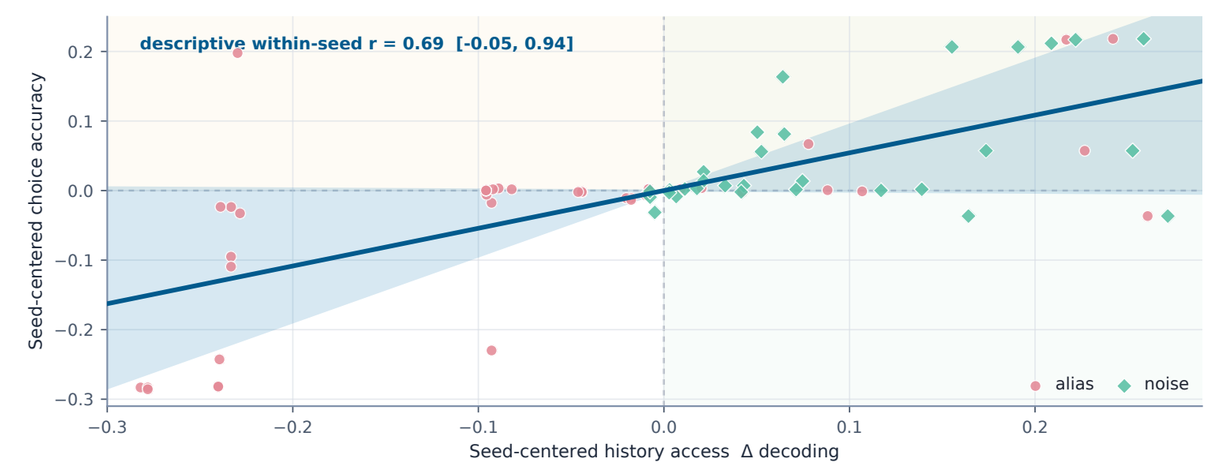
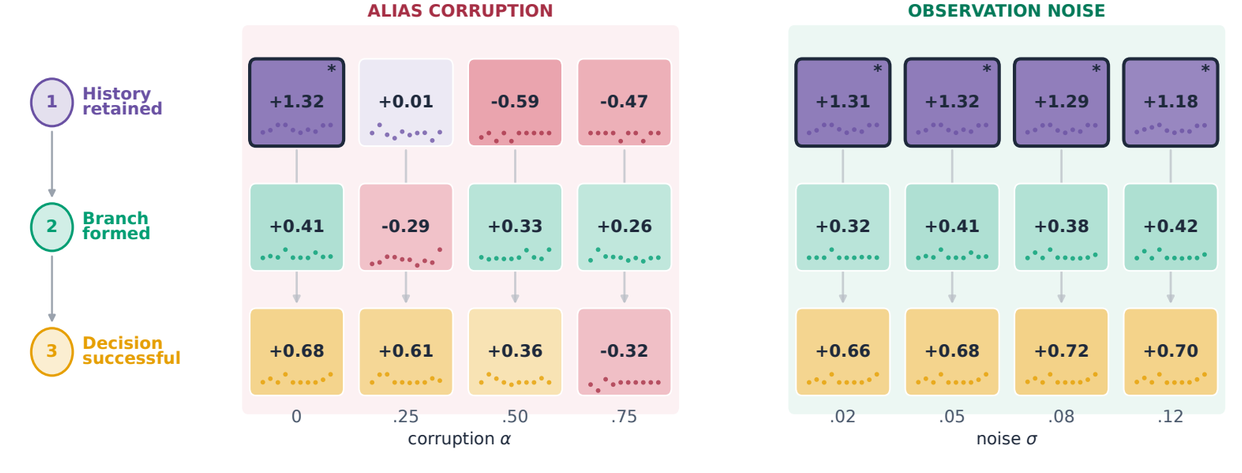

a
STRUCTURAL ALIASING
distinct cues
aliased cue
OBSERVATION NOISE
clean cue
noisy cue
same frozen checkpoint
EC
CA3
CA1
choice


b
alias branch
observation-noise branch
PAM
violet halo: history-access strength
Behavioral accuracy
0.70
0.65
0.60
0.55
0.50
α = .25
α = .50
α = .75
σ = .12
σ = .08
σ = .02
0.50
0.55
0.60
0.65
0.70
0.75
0.80
History decoding (frozen transfer probe)
seed distribution of behavior


c
HDGM
PAM
stateless
Alias α
Noise σ
Seed
101
202
303
404
505
606
707
808
909
1010
0 .25 .50 .75
.02 .05 .08 .12
Mean accuracy
Zoom: 0.47–0.56
Choice accuracy
0.0
0.5
1.0


d
Alias corruption
HDGM-PAM AUC · Holm p=0.0078
0
.25
.50
.75
0.0
0.2
0.4
History access Δ decoding · corruption α
Observation noise
HDGM-PAM AUC · Holm p=0.0078
.02
.05
.08
.12
0.0
0.2
0.4
History access Δ decoding · noise σ


e
descriptive within-seed r = 0.69 [-0.05, 0.94]
Seed-centered choice accuracy
Seed-centered history access Δ decoding
alias
noise


f
1 History retained
2 Branch formed
3 Decision successful
ALIAS CORRUPTION
+1.32
+0.01
-0.59
-0.47
+0.41
-0.29
+0.33
+0.26
+0.68
+0.61
+0.36
-0.32
0
.25
.50
.75
corruption α
OBSERVATION NOISE
+1.31
+1.32
+1.29
+1.18
+0.32
+0.41
+0.38
+0.42
+0.66
+0.68
+0.72
+0.70
.02
.05
.08
.12
noise σ